

Notes: Campbell (JF, 2006)

Household Finance

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1 Big picture

- **Question.** What should the field of **household finance** study, and how should we compare what households *actually do* with what they *should do* according to normative finance?
- **Setting.** This paper is a broad survey and research agenda rather than a single structural estimation paper. Campbell organizes the field around three central household decisions: **participation and asset allocation, diversification, and mortgage choice.**
- **Core challenge.** Household finance is hard because:
 - household data are difficult to measure,
 - households face long horizons, human capital risk, housing, borrowing constraints, and taxes,
 - many observed choices may reflect either **small frictions** or genuine **investment mistakes.**
- **Main conceptual contribution.** The paper defines household finance as the comparison between:
 - **positive household finance:** what households do, and
 - **normative household finance:** what they should do in well-specified models.

The gap between the two is the core object of interest.

- **Why this matters.** Household decisions directly affect welfare. A mistake in portfolio choice, diversification, or mortgage refinancing may have only limited effect on asset prices, but can have a **large effect on the household itself.**
- **Main substantive claim.** Many households make decisions that are close enough to optimal once one allows for realistic frictions, but a **minority** make important mistakes. These mistakes are more common among **poorer and less educated households.**
- **Main empirical themes.**
 - Many households do not participate in risky asset markets.
 - Many who do participate hold underdiversified portfolios.
 - Many households fail to refinance mortgages when refinancing is clearly attractive.
 - Some households appear to understand their limits and avoid financial strategies they are poorly equipped to manage.
- **Main equilibrium theme.** Retail financial innovation can be slowed by **advertising/education costs** and by **cross-subsidies from naive to sophisticated households** embedded in existing products.
- **Economic takeaway.** Household finance is not just asset pricing applied to consumers. It is a field about **mistakes, product design, regulation, disclosure, default options, and financial education.** The central policy goal is to improve what Campbell calls **financial hygiene.**

2 Data and measurement (key objects)

2.1 Measurement challenge: what an ideal dataset would look like

- Campbell argues that an ideal positive-household-finance dataset would have five features:
 1. representative coverage of the entire population,
 2. full measurement of total wealth,
 3. fine disaggregation across asset classes and ideally individual securities,
 4. high reporting accuracy,
 5. panel structure over time.
- The core problem is that household finances are private, scattered across multiple accounts, and often too complicated for survey respondents to report accurately.

2.2 Observed data sources used in the paper's discussion

- **Survey of Consumer Finances (SCF).** Best broad U.S. survey for wealth and asset allocation. Strong on coverage and total wealth, weaker on individual-security detail and panel structure.
- **Health and Retirement Study (HRS).** Useful especially for older households; range reporting helps with nonresponse.
- **Brokerage-account data.** Very accurate at the account level and useful for trading and direct-stock holdings, but not representative of total household wealth.
- **401(k) and retirement-account data.** Useful for inertia, defaults, and asset allocation inside retirement plans, but incomplete for total household portfolios.
- **Centralized share registers and tax records.** Examples include Finnish securities data and U.S. tax data.
- **Swedish administrative wealth data.** A uniquely rich panel with almost full-population coverage, individual-security holdings, real estate, and demographics.
- **American Housing Survey (AHS).** Used for mortgage rates and refinancing behavior.

2.3 Key descriptive objects from participation and asset allocation

- In the 2001 SCF, the **median household** has roughly:
 - \$35,000 in financial assets,
 - \$86,000 in net worth,
 - \$135,000 in total assets.
- Wealth is extremely skewed, so aggregate asset demands are heavily influenced by the wealthy.
- Poorer households mostly hold **liquid assets and vehicles**; middle-class households are dominated by **owner-occupied housing**; wealthier households hold much more **public equity and private business equity**.

- Even among relatively wealthy households, nonparticipation is not negligible. Around the **80th percentile of the wealth distribution**, roughly **20%** of households still own no public equity.
- Private business assets are an important substitute for public equity in wealthy portfolios.

2.4 Key descriptive objects from diversification evidence

- In Sweden, about **62%** of households participate in financial markets beyond bank deposits and money market funds.
- The median risky portfolio has a standard deviation of about **20%**.
- For the median Swedish household, **more than half of risky portfolio variance is idiosyncratic**.
- The median household's diversification mistake is modest in dollar terms—a little above **1%** of risky return or roughly **\$100 per year** relative to the currency-hedged world index under the CAPM benchmark.
- But the right tail is important: the **95th percentile** of return loss is above **\$2,200 per year** relative to the hedged world index.

2.5 Key descriptive objects from mortgage data

- Fixed-rate mortgages (FRMs) dominate the U.S. market, accounting for about **72%** of newly issued mortgages on average from 1985 to 2005.
- Time variation in FRM versus ARM use suggests that households respond not only to the FRM–ARM spread, but also to the **recent change in long-term mortgage rates**, which is hard to reconcile with a fully rational benchmark.
- In the 2003 AHS, after a large decline in fixed mortgage rates, more than:
 - **half** of households were paying more than **1%** above the current rate,
 - more than **one-third** were paying more than **1.5%** above,
 - about **one-quarter** were paying more than **2%** above,
 - about **one-eighth** were paying more than **3%** above.
- Almost **20% of mortgage dollars** were paying spreads above **2%** in 2003.

2.6 Measurement limits and interpretation

- **Age, time, and cohort effects** are fundamentally hard to separate in cross-sections.
- Education is likely endogenous: it may proxy for cognitive ability, financial literacy, or lower objective participation costs.
- Brokerage records reveal trading behavior, but not complete household portfolios.
- Swedish administrative data are unusually powerful, but external validity is not automatic.

- A central interpretive issue throughout the paper is whether observed nonstandard behavior reflects **preferences** or **mistakes**.

3 Models and empirical specifications (all equations + interpretation)

3.1 Positive versus normative household finance

- **Positive household finance** asks how households actually use financial instruments.
- **Normative household finance** asks how households should choose portfolios, mortgages, and other contracts in models with long horizons, labor income, housing, borrowing constraints, and taxes.
- Campbell's key conceptual move is to treat the difference between the two as the natural place to study **investment mistakes**.

3.2 Normative household portfolio problem: why textbook finance is incomplete

- The normative benchmark is an intertemporal portfolio problem in the spirit of **Merton (1971, 1973)** rather than a one-period mean–variance problem.
- The state variables that matter include:
 - expected returns and real interest rates,
 - nontraded human capital,
 - owner-occupied housing,
 - borrowing constraints,
 - taxes,
 - finite life-cycle horizon.
- A key qualitative implication is that asset demand is a **hedging problem**: households should care not just about shocks to wealth, but also about shocks to future investment opportunities and borrowing conditions.

3.3 Diversification metrics in the Swedish administrative data

If household h has risky-asset portfolio weight vector ω_h and the estimated variance–covariance matrix of risky returns is Σ , the risky portfolio variance is

$$\text{Var}(R_h) = \omega_h' \Sigma \omega_h.$$

Interpretation. This is the direct portfolio-risk measure made possible by household-level administrative holdings data. It captures the risk of the household's risky portfolio at a point in time.

Under a stylized symmetric decomposition, idiosyncratic portfolio variance is written as

$$\sigma_{ih}^2 = C_{ah}\sigma_{ah}^2 + (1 - C_{ah})\rho_{ah}\sigma_{ah}^2,$$

where $C_{ah} = \omega_h' \omega_h$ measures concentration, σ_{ah}^2 is average asset-specific idiosyncratic variance, and ρ_{ah} is average correlation across assets.

Interpretation. Idiosyncratic risk comes from three sources: volatile assets, concentrated holdings, and correlation among assets. In the Swedish data, the main problem is not correlation; it is concentrated holdings of one or two volatile directly owned stocks.

A log-linearized decomposition is

$$\log(\sigma_{ih}) \approx \log \sigma_{ah} + \frac{1}{2} \log C_{ah} + \frac{1}{2} \left(\frac{1}{\bar{C}_a} - 1 \right) \rho_{ah}.$$

If mutual funds are assumed fully diversified, an alternative decomposition is

$$\log(\sigma_{ih}) \approx \log D_h + \log \sigma_{sh} + \frac{1}{2} \log C_{sh} + \frac{1}{2} \left(\frac{1}{\bar{C}_s} - 1 \right) \rho_{sh},$$

where D_h is the share of directly held stocks in the risky portfolio.

Interpretation. This version makes clear that a large share of directly held stocks—rather than mutual funds—is a first-order source of household underdiversification.

Campbell then discusses three inefficiency measures. One is the **Sharpe ratio loss**,

$$1 - \frac{S_h}{S_B},$$

where S_h is the household Sharpe ratio and S_B is the benchmark Sharpe ratio.

A second is the **return loss** at the household's chosen risk,

$$w_h(S_B\sigma_h - \mu_h),$$

where w_h is the share invested in risky assets, σ_h is risky portfolio volatility, and μ_h is the household portfolio's mean risky return.

A third is the utility-equivalent riskless-rate loss,

$$\frac{w_h\sigma_h(S_B^2 - S_h^2)}{2S_h}.$$

Interpretation. These measures rank how costly underdiversification is. Median losses are modest, but the right tail is economically important.

Under the CAPM benchmark, Campbell writes overall return loss as

$$w_h(S_B\sigma_h - \mu_h) = w_h\beta_h^e \left(\frac{S_B}{S_h} - 1 \right) Er_m^e,$$

where β_h^e is the beta of risky holdings with the benchmark portfolio and Er_m^e is the expected excess return on the world market portfolio.

Interpretation. Household loss depends on three household-specific components: how aggressively the household invests, how much systematic market exposure it takes, and how inefficient its risky portfolio is relative to the benchmark.

3.4 Mortgage choice model from Appendix A

Campbell's simplified mortgage model compares an adjustable-rate mortgage (ARM) with a fixed-rate mortgage (FRM).

For an ARM, the nominal rate is state contingent:

$$R_{AH} = (1 + r)(1 + \pi_H) - 1 \approx r + \pi_H,$$

$$R_{AL} = (1 + r)(1 + \pi_L) - 1 \approx r + \pi_L.$$

Interpretation. The ARM is a floating-rate liability. If inflation is high, nominal mortgage payments rise immediately.

The date-1 real payment under an ARM is

$$\frac{R_{Ai}}{1 + \pi_i} M = \left(r + \frac{\pi_i}{1 + \pi_i} \right) M, \quad i \in \{H, L\},$$

and the date-2 real payment is

$$\frac{1 + R_{Ai}}{(1 + \pi_i)^2} M = (1 + r) \left(1 - \frac{\pi_i}{1 + \pi_i} \right) M.$$

Interpretation. Total discounted real payments are unchanged, but their timing depends on inflation. Higher inflation forces faster real repayment early on.

For an FRM, the real payment at date 1 is

$$\frac{R_F}{1 + \pi_i} M,$$

and the date-2 real payment is

$$\frac{1 + R_F}{(1 + \pi_i)^2} M.$$

Interpretation. A nominal FRM benefits the borrower in high-inflation states because the real burden of nominal debt falls. This is the key reason an FRM can be valuable even though a floating-rate note is safer in textbook bond-pricing logic.

Why FRMs can be optimal for households

- If households can borrow and lend freely and lenders are risk neutral, the **ARM** is preferred because lifetime resources are the same on average and the ARM avoids inflation risk priced into the FRM.
- If households are **borrowing constrained**, the comparison changes. The FRM back-loads payments and lowers the volatility of early consumption, which can dominate.
- Thus, a borrowing-constrained, risk-averse household may prefer an FRM even though a textbook investor would regard a long nominal bond as risky.

Refinancing option

A refinanceable FRM gives the borrower the right to replace the old loan when rates fall. This option reduces downside payment risk but raises the contract's initial rate.

Interpretation. The refinancing option is valuable, but households often fail to use it optimally. This is one of Campbell's central examples of a household financial mistake.

3.5 Equilibrium in retail financial markets: cross-subsidies and slow innovation

Suppose a fraction α of households is naive and the rest are sophisticated. For an existing product with total cost C_0 , costs across the two groups satisfy

$$\alpha C_0^N + (1 - \alpha)C_0^S = C_0.$$

Define the cross-subsidy from naive to sophisticated households as

$$x = C_0^N - C_0^S.$$

Then

$$C_0^S = C_0 - \alpha x.$$

Interpretation. Sophisticated households enjoy lower effective costs because naive households fail to exploit options or avoid hidden costs.

Now suppose a new product lowers production cost by g , so that

$$C_1 = C_0 - g.$$

Sophisticated households switch only if the innovation gain exceeds the lost cross-subsidy:

$$g > \alpha x.$$

Interpretation. Even a socially better product may fail to diffuse because sophisticated consumers lose the subsidy they were receiving from naive consumers in the incumbent product.

With advertising cost k and switching cost s , adoption requires

$$(1 - \alpha)(g - \alpha x - s) > k.$$

Interpretation. Retail financial innovation can be privately unattractive even when it is socially valuable. This is Campbell's equilibrium explanation for the slow adoption of products like automatically refinancing or inflation-indexed mortgages.

4 Identification and instruments (what makes the estimates credible)

4.1 What standard textbook finance misses

- One-period mean-variance theory misses life-cycle planning, human capital, housing, borrowing constraints, taxes, and limited financial sophistication.

- Observed household mistakes therefore cannot be judged against a frictionless one-period benchmark.

4.2 What identifies the main empirical patterns

- **Participation and broad asset allocation** are identified using the SCF, which has broad wealth coverage and detailed balance-sheet information.
- **Diversification** is identified much more cleanly in Swedish administrative data because the researcher observes holdings down to the individual security and can compute portfolio risk directly.
- **Mortgage refinancing sluggishness** is identified using household-level mortgage-rate distributions in the AHS, together with regressions that control for origination-year effects and include measures of collateral and income constraints.
- The paper builds credibility not from a single causal experiment, but from the fact that the same mistake pattern appears in several settings and lines up with demographic proxies for sophistication.

4.3 Why the mistake interpretation is plausible

- Mistakes are systematically more common among poorer and less educated households.
- Some households appear aware of their own limitations and avoid markets in which they would likely diversify badly.
- Mortgage refinancing mistakes are not fully explained by collateral or income problems.
- The paper repeatedly stresses that not all nonstandard behavior is a mistake; some can be explained by small objective frictions.

4.4 Main limitations

- Age, time, and cohort effects cannot all be separately identified.
- Education may proxy unobserved ability, not just learned finance.
- Swedish administrative evidence is unusually strong, but external validity is not automatic.
- Brokerage-account evidence may overstate active trading or underdiversification because such accounts need not represent total wealth.

4.5 Relation to the literature

- Relative to **asset pricing**, the paper focuses on the typical household rather than the wealthy marginal investor.
- Relative to **corporate finance**, it studies how households use financial contracts to pursue their objectives.

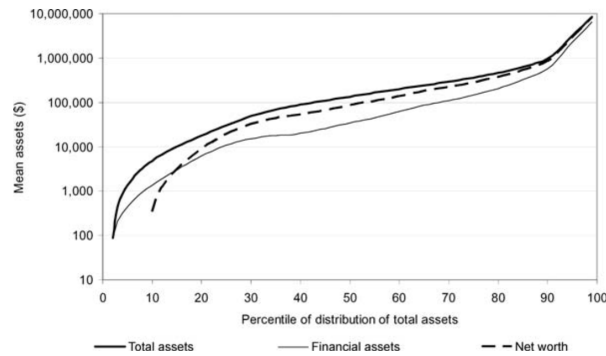


Figure 1. The U.S. wealth distribution. The cross-sectional distribution of total assets, financial assets, and net worth in the 2001 Survey of Consumer Finances.

- Relative to **behavioral finance**, Campbell takes the view that standard finance remains the welfare benchmark and that deviations are often mistakes rather than merely different preferences.
- The paper also launches a policy-oriented agenda that Campbell calls **household financial engineering**.

5 Tables and figures

5.1 Figure 1: U.S. wealth distribution

- **Read:** The figure plots total assets, financial assets, and net worth by percentile of the household wealth distribution in the 2001 SCF.
- **Main message:** Financial assets are tiny for much of the population. The median household has only about \$35,000 in financial assets.
- **Why it matters:** Typical households are very different from the wealthy households that dominate aggregate asset demand. This is one reason household finance is distinct from standard asset pricing.

5.2 Figure 2 and Figure 3: participation and portfolio shares across wealth

- **Figure 2.** Participation in risky asset classes rises strongly with wealth, but is far from universal even among affluent households.
- **Figure 3.** Portfolio composition changes systematically with wealth:
 - poor households: liquid assets and vehicles,
 - middle class: real estate dominates,
 - rich households: equity and private business assets dominate.
- **Main message:** Wealth affects both participation and conditional asset shares. Fixed costs and background risk are natural interpretations.

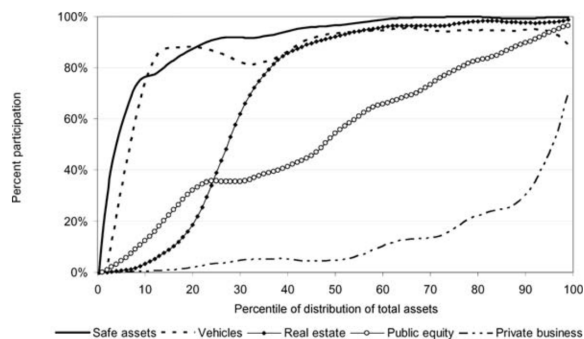


Figure 2. Participation rates by asset class. The cross-sectional distribution of asset class participation rates in the 2001 Survey of Consumer Finances.

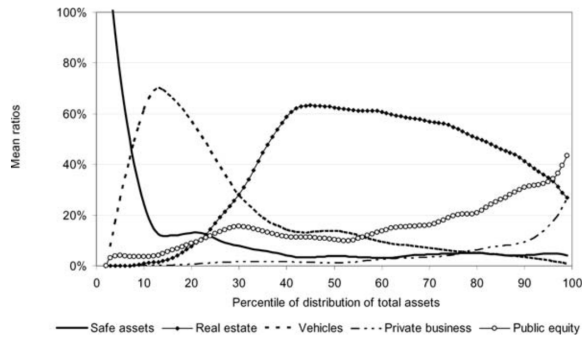


Figure 3. Asset class shares in household portfolios. The share of each asset class in the aggregate portfolio of households at each point in the wealth distribution, in the 2001 Survey of Consumer Finances.

Table I
Equity Participation and Portfolio Share

The table reports demographic determinants of participation in public equity and private business ownership (logit regressions, left panel) and of the portfolio share among participants (OLS regressions, right panel) for households in the 2001 Survey of Consumer Finances. Standard errors are reported underneath the coefficients in parentheses. Coefficients significant at the 10% level are denoted by *, at the 5% level by **, and at the 1% level by ***. All regressions control for gender and marital status. In the reference household, the household head is a married white male with no high school diploma. The column headed "Probability Estimates" reports the probability of participation for the reference household, and the change in this probability caused by a unit change in a binary variable and a one-standard-deviation change in a continuous variable.

Dependent Variable	Whether a Household Owns a Given Asset				Portfolio Share for Participants	
	Public Equity		Private Business		Public Equity	Private Business
	Coefficients	Probability Estimates (%)	Coefficients	Probability Estimates (%)	Coefficients	Coefficients
<i>Reference household</i>		54.5		0.7		
No tolerance for investment risk	-1.173*** (0.121)	-40.9	0.189 (0.147)	0.5	-0.032** (0.014)	0.061* (0.032)
Age	-0.034* (0.019)	-20.2	0.094*** (0.031)	-0.6	0.002 (0.002)	-0.003 (0.006)
Age squared	3.9×10^{-4} (1.8×10^{-4})		-1.9×10^{-5} *** (3.0×10^{-5})		-1.7×10^{-5} (2.4×10^{-5})	-5.6×10^{-6} (7.6×10^{-6})
White	0.133 (0.128)	5.3	0.335* (0.138)	0.4	0.032** (0.014)	0.009 (0.038)
High school diploma	0.401*** (0.157)	14.9	-0.086 (0.236)	-0.2	0.009 (0.020)	0.028 (0.049)
College diploma	0.848*** (0.186)	27.9	-0.211 (0.275)	-0.3	0.037* (0.021)	-0.041 (0.052)
Graduate school	1.051*** (0.239)	32.2	-0.210 (0.280)	-0.3	0.040* (0.023)	-0.038 (0.051)
Number of children	-0.010 (0.049)	-0.5	-0.068 (0.058)	-0.1	-0.011* (0.005)	0.003 (0.011)
Ln(income)	2.650* (1.452)	17.3	-2.507** (1.036)	-0.4	0.038 (0.153)	0.196 (0.283)
Ln(income) squared	-0.101 (0.070)		0.101** (0.047)		-0.002 (0.007)	-0.007 (0.012)
Ln(wealth)	0.094 (0.372)	37.3	0.315 (0.583)	30.6	-0.312*** (0.047)	-0.409*** (0.120)
Ln(wealth) squared	0.026 (0.017)		0.029 (0.023)		0.014*** (0.002)	0.013*** (0.005)
Sample size	4,304		4,304		2,822	1,845

5.3 Table I: demographic determinants of participation and portfolio shares

- **Read:** The left panel studies whether households hold public equity or private business assets; the right panel studies conditional shares among participants.
- **Key findings for public equity.**
 - Strong positive effects of **education**, **income**, and **wealth** on participation.
 - Weak negative age effect once wealth and income are controlled for.
 - Households that report no tolerance for investment risk are less likely to hold equity.
- **Key findings for private business.**
 - Ownership is concentrated among high-income and high-wealth households.
 - Age effects are hump-shaped.
- **Main message:** Observable sophistication variables strongly predict who enters risky asset markets.

5.4 Administrative-data diversification evidence from Sweden

- **Read:** Swedish administrative data let Campbell, Calvet, and Sodini directly compute household portfolio variance and separate systematic from idiosyncratic risk.

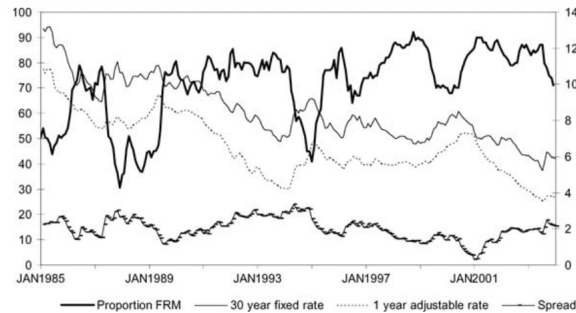


Figure 4. Fixed rate mortgage share and mortgage rates. The share of newly originated mortgages with fixed rates, reported in the monthly survey of the Federal Housing Finance Board, together with 30-year fixed and 1-year adjustable mortgage rates reported by the Federal Home Loan Mortgage Corporation (FHLMC) and the spread between them, from 1985 to 2005.

- **Main findings.**
 - The median risky portfolio volatility is about 20%.
 - More than half of median portfolio variance is idiosyncratic.
 - Median welfare loss from underdiversification is modest, but the right tail is large.
- **Economic interpretation:** Underdiversification is not a universal disaster, but it is a serious problem for a minority of households.

5.5 Figure 4: fixed-rate mortgage share and mortgage rates

- **Read:** The figure plots the share of newly originated FRMs together with ARM and FRM rates and their spread from 1985 to 2005.
- **Main message:** Households appear more likely to choose ARMs after recent increases in FRM rates and more likely to choose FRMs after recent declines.
- **Why it matters:** This pattern is hard to reconcile with a fully rational benchmark, because recent changes in long-term rates should matter only insofar as they predict future rates.

5.6 Figure 5 and Table II: mortgage spreads and refinancing behavior

- **Figure 5.** The distribution of mortgage spreads shifts far to the right in 2003 after a sharp fall in mortgage rates. Many households keep paying rates far above current market rates.
- **Table II.** Refinancing between 2001 and 2003 is more common among **younger, smaller, better educated, higher-income, white households with more expensive homes**.
- **Interpretation:** Some sluggish refinancing reflects collateral constraints, but the demographic pattern strongly suggests a role for **financial sophistication**.

5.7 Section V mechanism: hidden costs and cross-subsidies

- **Read:** Campbell's simple model of naive and sophisticated consumers explains why better financial products may fail to diffuse.

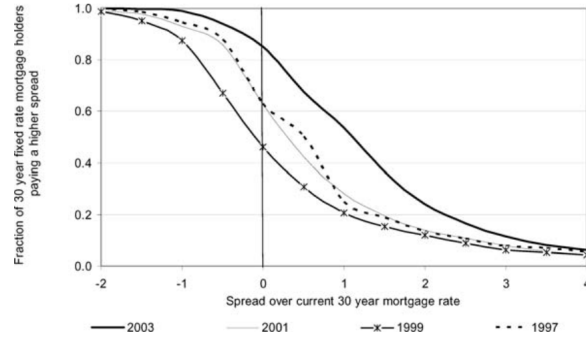


Figure 5. Distribution of mortgage spreads. The distribution of self-reported mortgage rates, corrected for implausibly low rates, in the 1997, 1999, 2001, and 2003 American Housing Surveys, relative to the currently prevailing FHLMC rate in the year of each survey.

Table II
Mortgage Refinancing and Mortgage Rates

The table reports demographic determinants of mortgage behavior for households with 30-year fixed mortgages in the 2001 and 2003 American Housing Surveys. The first three columns report probit regressions of refinancing between 2001 and 2003, confirmed changes of address between 2001 and 2003, and 2001 self-reporting of implausibly low mortgage rates more than 2% below standard FHLMC rates prevailing at origination, onto demographic variables. The last two columns report OLS regressions of self-reported 30-year fixed mortgage rates in 2001 and 2003, adjusted to correct for implausibly low self-reported rates, onto demographic variables. Standard errors are reported underneath the coefficients in parentheses. Coefficients significant at the 10% level are denoted by *, at the 5% level by **, and at the 1% level by ***. All regressions include year and region fixed effects, and control for gender, marital status, and urban house location. In the reference household, the household head is a married white male with no high school diploma. The column headed "Probability Estimates" reports the probability for the reference household, and the change in this probability caused by a unit change in a binary variable and a one-standard-deviation change in a continuous variable.

Dependent Variable	Whether a Household Refinanced a 30-year Fixed Mortgage in 2001–2003		Whether a Household Moved between 2001 and 2003		Whether a Household Reported a Rate 2 or More % Points Below FHLMC in 2001		Mortgage Rate for 30-year Fixed Mortgages in 2001		Mortgage Rate for 30-year Fixed Mortgages in 2003	
	Coefficients	Probability Estimates (%)	Coefficients	Probability Estimates (%)	Coefficients	Probability Estimates (%)	Coefficients	Probability Estimates (%)	Coefficients	Probability Estimates (%)
Reference household		27.9		4.5		1.3		0.097**		0.060**
Loan problem	–0.179*** (0.071)	–4.6	–0.050 (0.088)	–0.5	0.233*** (0.082)	2.2	0.097** (0.048)	0.060** (0.031)		
Income problem	–0.076 (0.099)	–0.2	0.132 (0.130)	1.4	–0.064 (0.107)	–0.6	0.139** (0.063)	0.364*** (0.065)		
First house	0.104** (0.043)	3.6	0.001 (0.054)	0.0	–0.037 (0.053)	0.2	–0.039 (0.035)	–0.057** (0.029)		
Age	–0.028** (0.011)	–4.3	–0.078*** (0.012)	–1.6	0.003 (0.010)	–0.1	0.007 (0.008)	0.004 (0.007)		
Age squared	1.5×10^{-4} (1.1×10^{-4})		6.8×10^{-4} *** (1.3×10^{-4})		-2.3×10^{-5} (1.2×10^{-4})		-4.6×10^{-5} (9.2×10^{-5})	-4.1×10^{-6} (8.3×10^{-5})		
White	0.250*** (0.056)	7.7	0.291*** (0.081)	1.9	0.071 (0.071)	–0.1	0.002 (0.041)	–0.202*** (0.038)		

High school diploma	0.163** (0.077)	5.4	0.293*** (0.115)	3.5	–0.272*** (0.076)	–0.6	–0.156** (0.062)	–0.170*** (0.062)
College diploma	0.268*** (0.083)	9.2	0.364*** (0.122)	4.6	–0.203** (0.091)	–0.5	–0.095 (0.067)	–0.180*** (0.055)
Graduate school	0.261*** (0.091)	8.9	0.316** (0.132)	3.9	–0.296*** (0.104)	–0.7	–0.070 (0.074)	–0.154** (0.063)
Number of children	–0.006 (0.017)	–0.3	–0.060** (0.024)	–0.6	2.5×10^{-4} (0.028)	0.1	0.006 (0.013)	0.004 (0.011)
Number of adults	–0.094*** (0.028)	–2.6	–0.124*** (0.041)	–0.8	–0.006 (0.031)	–0.1	–0.015 (0.022)	0.003*** (0.020)
Ln(income)	0.274** (0.140)	1.3	–0.528** (0.224)	0.9	0.049 (0.107)	–0.3	–0.037 (0.110)	0.018 (0.096)
Ln(income) squared	–0.011* (0.007)		0.028*** (0.011)		–0.007 (0.006)		0.000 (0.005)	0.000 (0.005)
Ln(house value)	1.259* (0.747)	6.6	1.589* (0.960)	–0.2	1.422* (0.822)	0.1	1.101** (0.464)	–0.233 (0.440)
Ln(house value) squared	–0.042 (0.031)		–0.068* (0.040)		–0.058* (0.034)		–0.045** (0.019)	0.010 (0.018)
Ln(mortgage)	–0.024 (0.136)	–0.1	–0.024 (0.045)	0.0	–0.057* (0.031)	0.0	0.335*** (0.079)	0.447*** (0.054)
Ln(mortgage) squared	0.003 (0.007)		0.002 (0.002)		–0.003* (0.002)		–0.027*** (0.004)	–0.038*** (0.003)
Sample size	5,190		5,735		7,610		7,610	9,749

- **Main message:** Sophisticated consumers may stay with an inferior incumbent product because leaving it means giving up a cross-subsidy funded by naive consumers.
- **Policy relevance:** This helps explain slow innovation in mortgages and other retail financial products even in competitive markets.

6 Short summary

- **Method.** A conceptual and empirical survey that compares positive household behavior with normative portfolio and mortgage theory.
- **Data.** SCF and related U.S. surveys for asset allocation, Swedish administrative data for diversification, and AHS data for mortgages.
- **Main result.** The central problems in household finance are not universal irrationality, but important mistakes by a minority of households—especially nonparticipation, underdiversification, and refinancing failures.
- **Policy implication.** Financial education, better disclosure, suitable default options, and public support for simple well-designed financial products may raise welfare.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- Campbell defines household finance as the study of how households use financial instruments and where their actual decisions diverge from normative benchmarks.
- The paper shows that these divergences are especially visible in risky-market participation, diversification, and mortgage choice.
- The evidence suggests that a minority of households make serious mistakes, and these mistakes matter for welfare and policy even if aggregate prices are mostly set by sophisticated investors.

7.2 Economic intuition

- Household problems are hard: people face long horizons, labor income risk, illiquid housing, borrowing constraints, and taxes.
- Because the decision problem is hard, sophistication is scarce and valuable.
- Some households rationally avoid tasks they are not equipped to perform, but others remain in the market and make costly mistakes.
- These mistakes can persist because financial products themselves are designed in equilibrium, sometimes in ways that reward sophistication and exploit inattention.

7.3 Takeaway

Household finance should be understood as a field about **welfare, mistakes, and market design**. The key lesson of the paper is that improving household outcomes requires more than better theory of asset prices; it requires better **data**, better **consumer finance theory**, and better **institutions**—including disclosure, defaults, regulation, and financial products designed for ordinary households rather than just sophisticated ones.